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Diversity, Distribution and Species composition of Odonates in Ranchi, Jharkhand

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Abstract : Biodiversity protection and conservation is on national and international agenda and responsible for sustainable development of a region or a country. Therefore Odonates diversity in Ranchi District was observed during 2011- 2012 and 33 species belonging to 8 families were recorded. Altogether randomly collected 24 dragonflies species were sampled and the number of damselflies species samples were 09. Libellulidae was the richest family with 18 species and sub- order Anisoptera. Different habitat types provide good opportunities to these wonderful groups to flourish and survive. Mostly Odonates were aggregated due to habitat specific nature and random distribution indicates availability of resource utilization to survive.

Key words: Key Words- Odonata, Diversity, Libellulidae, Anisoptera, sustainable development,

INTRODUCTION

Dragon and Damselflies are amongst the most attractive creatures of the earth. Dragon and damselflies are collectively called Odonates[1]. The order Odonata is one of the ancient groups of insects. Dragonflies mostly occur in the vicinity of different freshwater habitats like rivers, streams, marshes, lakes and even small pools and rice fields[2]. Odonates are good indicators of environmental changes as they are sensitive to changes in the habitats, atmospheric temperature and the weather conditions[3,4]. They are biocontrol agents, many species of Odonates inhabiting agro ecosystems play an important role controlling pest populations (Tiple et al.2008)[5].

They spend a major part of their life cycle in fresh water ecosystem. The adults are generally predaceous insects, while the larvae are carnivorous and voracious. Even though the species are usually highly specific to a habitat, some have adapted to urbanization and use man-made water bodies. Being primarily aquati, their life history is closely linked to specific aquatic habitats (Andrew et al.2009)[2]. Fraser(1933[6],1934[7],1936[8]), Nesemann(2011)[9],Nair (2011)[10]

Emiliamma et al.(2002[11].), Srivastava(1997[12]),Mishra (2007[13]), Kumar & Mitra (2008[14]), and). Tiple et al.(2012[15]) worked a lot on Odonates.

There is no published checklist of different species Odonata of Ranchi is known. Hence, the present work was initiated.

MATERIALS AND METHODS

The Odonates were collected randomly from gardens, temporary and permanent water bodies, rivers, forests, agricultural fields, bushes, grasses, marshes, near contaminated water in drains and wherever seen. A suevey was undertaken from the year 2011- 2012.The adult specimen were identified with the help of identification keys provided by Fraser(1933,1934,1936),Mitra (2006),Subramanian(2005), Andrew et al. (2009), and Subramanian (2009).

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RESULTS & DISCUSSION

A total of 33 species of Odonata belonging to 8 families and two suborders were collected from the 16 selected area of Ranchi during the survey periods. The sampling of Odonates from the four Zones and 16 sites of Ranchi has resulted more no. of species dragonflies as compared to the number of species of damselflies. Altogether randomly collected 24 dragonflies species were sampled and the number of damselflies species samples were 09.

Among the 33 sp. Zone I recorded 20 sp., Zone II 18sp. Zone III 15 sp. And Zone IV 13 sp. On the basis of number of identified species of drgaonflies. *Libellilidae* was the most dominant family represented by 18 species followed by family *Aeshnidae* by 2 species family *Chlorogomphidae, Gomphidae, macroninae* represented by one species.

On the basis of number of identified species of damselflies, *Coenagrionidae* was the most dominant family represented by 6 species, followed by family *platycucnididae* by 2 species and *Lestidae* by one species.

Among dragonflies, belonging to sub order *Anisoptera,* the common sp. sampled from all the four

Table1: List of sampled Odonates of Ranchi city.

S. N.	Dragonflies Species (**Sub-order: Anisoptera**)	Family
1	*Acioma panoripoids,*Rombur, 1842	*Libellulidae*
2	*Aeshna sps.*Fabricius, 1775	*Aeshnidae*
3	*Aethriamanta brevipennis,* Rambur, 1842	*Libellulidae*
4	*Anex sps.*, Leach, 1815	*Aeshnidae*
5	*Brachithemis contaminata,* Fabricius, 1793	*Libellulidae*
6	*Chlorogomphus campioni,* Fraser, 1924	*Chlorogomphidae*
7	*Cratilla lineata,* Foerster, 1903	*Libellulidae*
8	*Crocothemis servilia,* Drury, 1770	*Libellulidae*
9	*Diplocodus trivialis,* Rambur, 1842	*Libellulidae*
10	*Gynacantha sps.,* Rambur, 1842	*Aeshnidae*
11	*Ictinogomphus rapax,* Rambur, 1842	*Gomphidae*
12	*Lathrecista asiatica* Fabricius, 1768	*Libellulidae*
13	*Macromia magnifica*.Kennedy,	*Macrominae*
14	*Orethrum pruinosum ,* Burmeister, 1839	*Libellulidae*
15	*Orthetrum glaucaum,* Brauer, 1865	*Libellulidae*
16	*Orthetrum Sabina*(Green Marsh Hawk), Drury, 1770	*Libellulidae*
17	*Pantala flavescens,* Fabricius, 1778	*Libellulidae*
18	*Potamarcha congener*Rambur, 1842	*Libellulidae*
19	*Rhyothemis variegata,* Linnaeus, 1763	*Libellulidae*
20	*Tramea basilaris,* Palisat de Beauvois, 1805	*Libellulidae*

Table1contd.....

Table1contd.....

21	*Trithemis aurora.* , Burmeister, 1879	*Libellulidae*
22	*Trithemis pallidinervis,* Kirby, 1889	*Libellulidae*
23	*Urothemis signata,* Rambur, 1842	*Libellulidae*
24	*Zyxomma petiolatum,* Rambur, 1842	*Libellulidae*

Damselflies

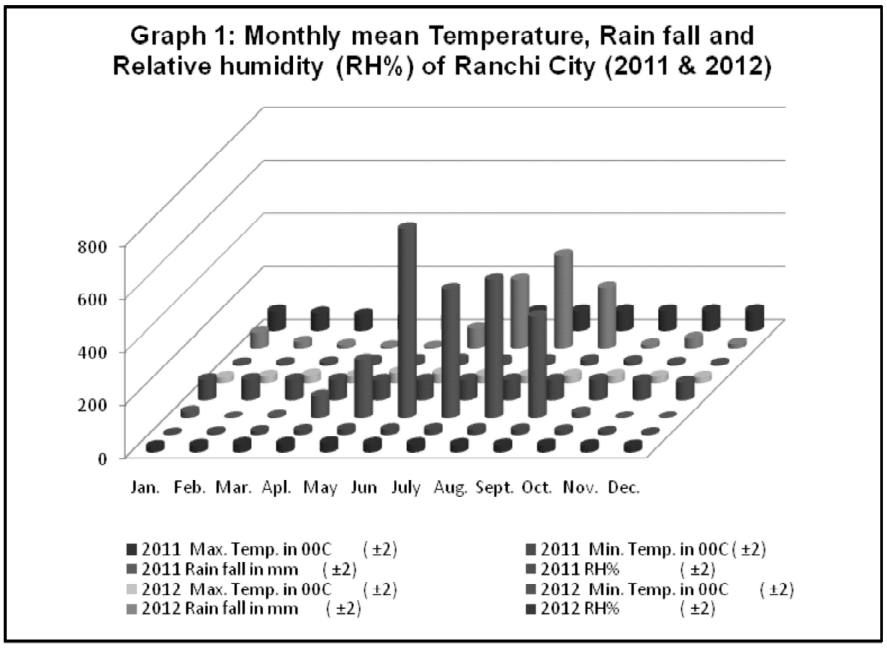
S. N.	Damselflies Species (**Sub-order: Zygoptera**)	Family
1	*Agriocnemis pygmaea*(Pigmy Dartlet), Rambur, 1842	Coenagrionidae
2	*Copera marginipes*(Yellow Bush Dart), Rambur, 1842	Platycnemididae
3	*Copera vittata* (Blue Bush Dart), Selys,1863	Platycnemididae
4	*Ischnura aurora* (Golden Dartlet), Brauer, 1865	Coenagrionidae
5	*Pseudagrion microcephalum* (Blue Grass Dartlet), Rambur, 1842	Coenagrionidae
6	*Ceriagrion olivaceum,* Laidlaw, 1914	Coenagrionidae
7	*Ceriagrion coromandelianum,* Fabricius,1798	Coenagrionidae
8	*Pseudagrion indicum,* Fraser, 1924	Coenagrionidae
9	*Lestes sps.,* (Spread wings) Leach, 1815	*Lestidae*

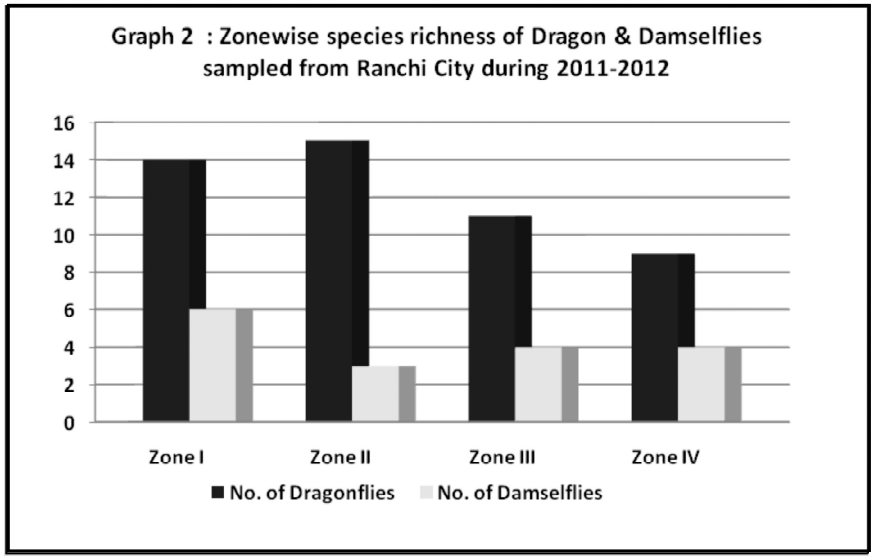
Table 2: Monthly mean Temperature, Rain fall and Relative humidity (RH%) of Ranchi City (2011 & 2012)

Months	2011				2012			
	Max. Temp. in 0^0C (±2)	Min. Temp. in 0^0C (±2)	Rain fall in mm (±2)	RH% (±2)	Max. Temp. in 0^0C (±2)	Min. Temp. in 0^0C (±2)	Rain fall in mm (±2)	RH% (±2)
January	21.8	4.5	23.8	78.90	21.2	8.3	60.1	76.50
February	26.6	9.3	0	79.80	25.6	9.2	25.2	68.75
March	32.7	14.5	6.3	78.75	30.8	12.3	12.8	61.30
April	33.4	18.3	82.4	78.65	25.9	19.0	7.5	56.70
May	35.8	21.9	218.1	76.90	38.7	21.3	6.0	54.90
June	32.6	21.3	714.3	77.10	36.2	22.7	75.4	63.85
July	31.5	22.4	486.2	78.75	30.4	20.2	260.4	74.15
August	29.4	21.7	522.6	78.25	29.4	20.6	351.5	76.90
September	28.2	20.5	382.6	77.75	30.0	20.6	230.1	77.30
October	28.1	17.4	20.4	78.20	29.0	17.2	14.3	76.95
November	25.5	11.0	0.0	75.65	24.9	10	39.0	77.20
December	23.5	5.8	0.0	68.25	21.9	6.6	17.4	77.55

Table 3: : Zonewise species richness of Dragon & Damselflies sampled from Ranchi City during 2011-2012

Zones	No. of Dragonflies	No. of Damselflies	Total
Zone I	14	6	20
Zone II	15	3	18
Zone III	11	4	15
Zone IV	9	4	13





zone were *Acioma panoripoids* Rambur (1842) and *Diplocodus trivialis* Rambur (1842), both representing the same family *Libellilidae.* The sp. *Crocothemis servilia* Drury (1770) belonging to the family *libellulidae* was found common to three zones viz. Zone I, Zone III & Zone IV.

The dragonfly species, *Orthetrum glaucaum* Brauer (1865) was also common to three zones viz. Zone I, Zone II & Zpme III. Drury (1770).

Similarly, from all these three zone the dragonfly O*rthtrum Sabina* Drury (1770) commonly called as Green Marsh Hawk was sampled as a common species. One more dragonfly species *Trithemis aurora* Burmeister (1879) from the family *Libellulidae* was collected from the three Zones viz. Zone II, Zone III & Zone IV.

Contrastingly in case of damselflies belonging to sub-order- Zygoptera, no species was found as a common species in all the four zones. But two species of damselflies viz. *Copera marginipes* Rambur (1842) generally termed as yellow Bush Dart , & *Copera vittata* Selys (1863) commonly called as Blue Bush Dart were common to the three zones. i.e. Zone I, II & IV and zone II, III & IV respectively.

The statistical indices indicates that the maximum diversity was observed at Kanke, one of the sites of Zone I, Ranchi i.e. North-West of (20 species) followed by Bariatu, the site of Zone II i.e. North – East of Ranchi city (18 species) of Hinoo, the site of Zone III i.e.South-East, Kokar, Namkum (15 sps.) and the site Zone IV i.e. south-West of Ranchi (13 sp). This could be attributed to the survey effort since at all these sites maximum survey were conducted (4 times). Kanke, Bariatu and the surrounding areas of Hinoo are of conservation importance since these sites hold a good diversity of Odonates.

CONCLUSION

Though the present study covers varied habitats in the Ranchi city, the authors are conscious of the fact that more extensive surveys are needed along the water bodies throughout the state to understand seasonality and flight periods.

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